Quantifying Post-Flood Recovery Dynamics in Sunamganj, Bangladesh: An Integrated Sentinel-1 SAR and Nighttime Light Analysis

1. Abstract

This study evaluates the impact and recovery of the catastrophic June 2022 floods in Sunamganj, Bangladesh. By integrating **Sentinel-1 SAR** imagery with **NASA VIIRS Night-time Light (NTL)** data and socio-economic indicators (Population Density, Relative Wealth Index, and Road Density), we mapped the flood extent and analyzed the factors influencing recovery speed. A linear regression model was applied to the flooded areas, revealing that **infrastructure (road density)** is the most significant predictor of recovery ($\beta_3 = 19.86$), while wealth and population density showed slight negative correlations with light-based recovery metrics.

2. Introduction

Sunamganj, located in the northeastern Haor basin of Bangladesh, is highly vulnerable to flash floods. In June 2022, record-breaking rainfall led to one of the most severe flooding events in recent history, displacing millions and destroying critical infrastructure. Traditional optical monitoring is often hindered by cloud cover during the monsoon; therefore, this study utilizes **Radar (SAR)** and **NTL** data to provide an unobstructed view of the disaster's physical extent and the subsequent socio-economic recovery.

3. Methodology

Study Area

The study focuses on Sunamganj District, located in northeastern Bangladesh, a region highly vulnerable to seasonal flooding

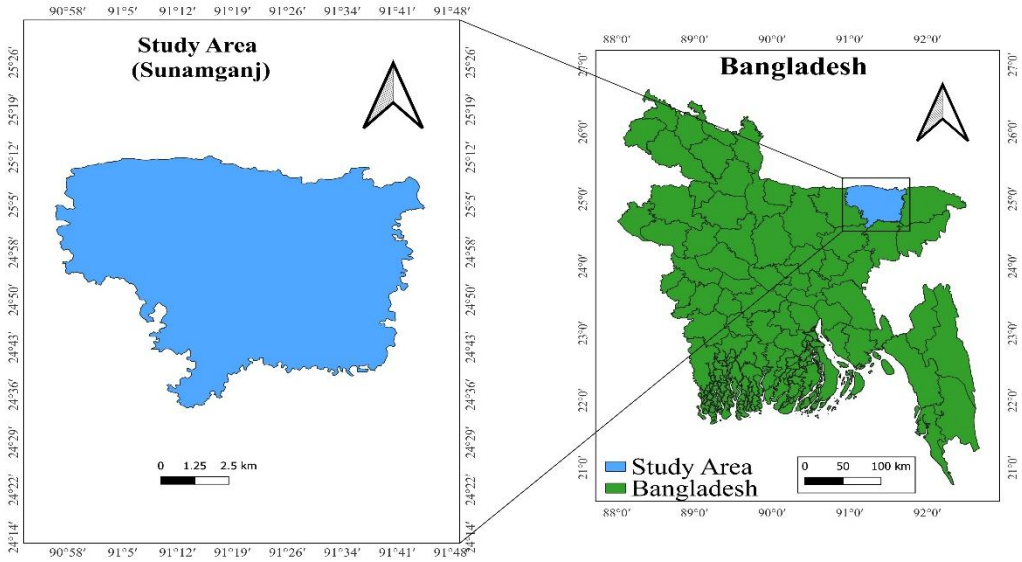

Figure 1: *Study Area Map*

The analysis was conducted in **Google Earth Engine (GEE)** using the following steps:

- **Flood Mapping:** Sentinel-1 Ground Range Detected (GRD) data in VV and VH polarizations were used to detect water. A "change detection" approach compared pre-flood (May 2022) and peak-flood (June 2022) backscatter.
- **Recovery Index:** Post-flood recovery was calculated using the **NTL Recovery Index**:

VIIRS VNP46A2 data were used to calculate mean NTL values for three periods:

- **Pre-flood (Baseline):** April 1 - May31, 2022.
- **Post-flood (Impact):** June 15 - July 5, 2022.
- **Recovery (Current):** January 1 – June, 2023

The index was calculated pixel-by-pixel as follows:

$$Y_{rec} = \frac{NTL_{Recovery} - NTL_{Post}}{NTL_{Pre} + 1}$$

Data Integration: The **Relative Wealth Index (RWI)** from Meta, **WorldPop** population data, and **JRC Global Human Settlement Layer (GHSL)** for roads were masked to only include the flooded pixels.

- **Statistical Analysis:** An **Ordinary Least Squares (OLS) Regression** was performed using the sampled points to quantify the relationship between recovery and socio-economic variables.

Multivariate Linear Regression

An Ordinary Least Squares (OLS) regression model was applied to determine the influence of the independent variables on the dependent NTL Flood Recovery Index. The model is specified as:

$$Y_{rec} = \beta_0 + \beta_1(PopDensity) + \beta_2(Wealth) + \beta_3(RoadDensity) + \varepsilon$$

4. Results

The regression model converged properly. The equation is:

$$Y_{rec} = 0.299 - 0.021(PopDensity) - 0.025\ (Wealth) + 19.865\ (RoadDensity) + \varepsilon$$

The regression analysis yielded the following coefficients:

Table 1: Regression Coefficient

Variable	**Coefficient (β)**	**Significance**
Intercept (β_0)	**0.299**	Baseline recovery across the region.
Road Density (β_3)	**19.865**	Dominant positive driver of recovery.
Wealth (RWI) (β_2)	**-0.025**	Slight negative correlation.
Pop Density (β_1)	**-0.021**	Slight negative correlation.

Visualizations:

Flood Extent Map: High concentrations of water were identified in the central and northern Haor regions. This map provides a comprehensive spatial overview of the inundation pattern across Sunamganj, one of Bangladesh’s most flood-prone districts. The blue areas indicate that a significant majority of the district's landmass was submerged during the 2022 event.

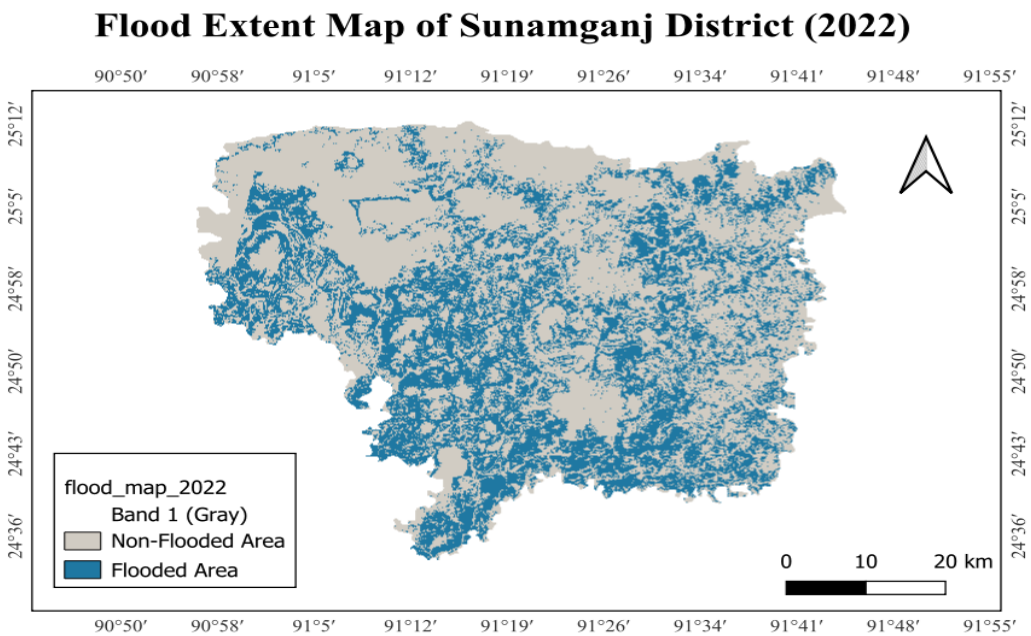

Figure 2: Flood Extent Map of Sunamganj District during the peak of the June 2022 disaster

Masked Flood Map: This map represents the functional input layer for the regression model, filtering out noise and focusing exclusively on the areas of impact.

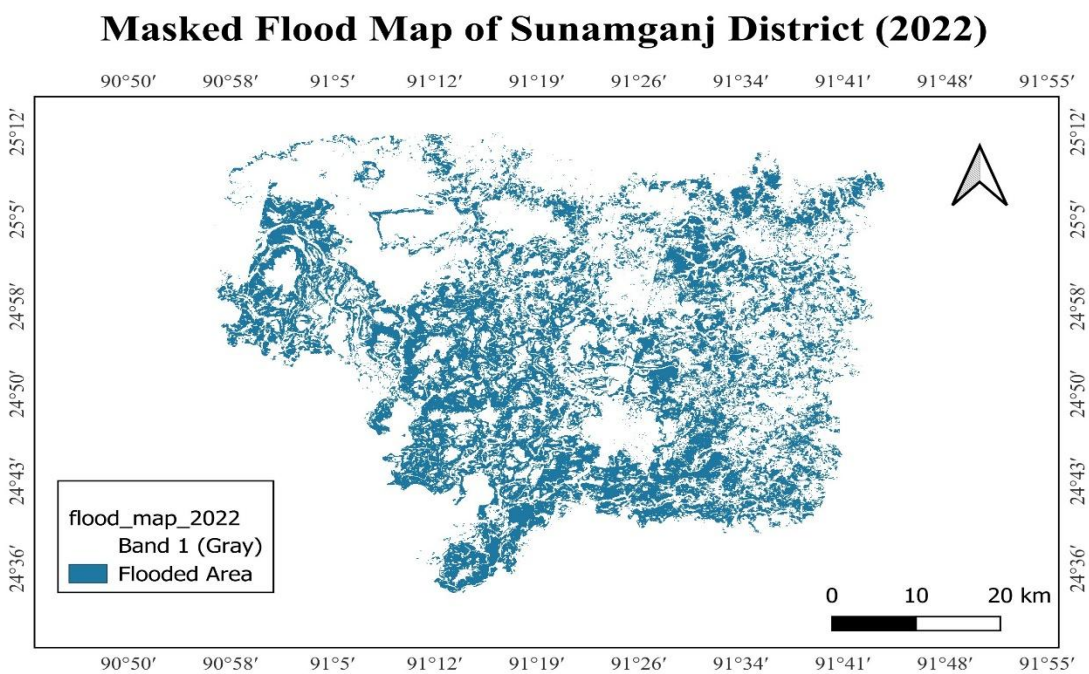

Figure 3: Masked Flood Extent Map of Sunamganj District (2022)

NTL Recovery Map: This map serves as the quantitative outcome layer of the study, illustrating the spatial distribution of the Nighttime Light (NTL) Recovery Index. It categorizes the flooded pixels based on their ability to return to pre-flood brightness levels. The map utilizes a "traffic-light" classification where **Red** indicates low recovery, **Yellow** represents moderate progress, and **Green** signifies areas that successfully restored or exceeded their pre-flood light intensity.

Recovery was strongest along major transport corridors and slightly slower in dense urban centers.

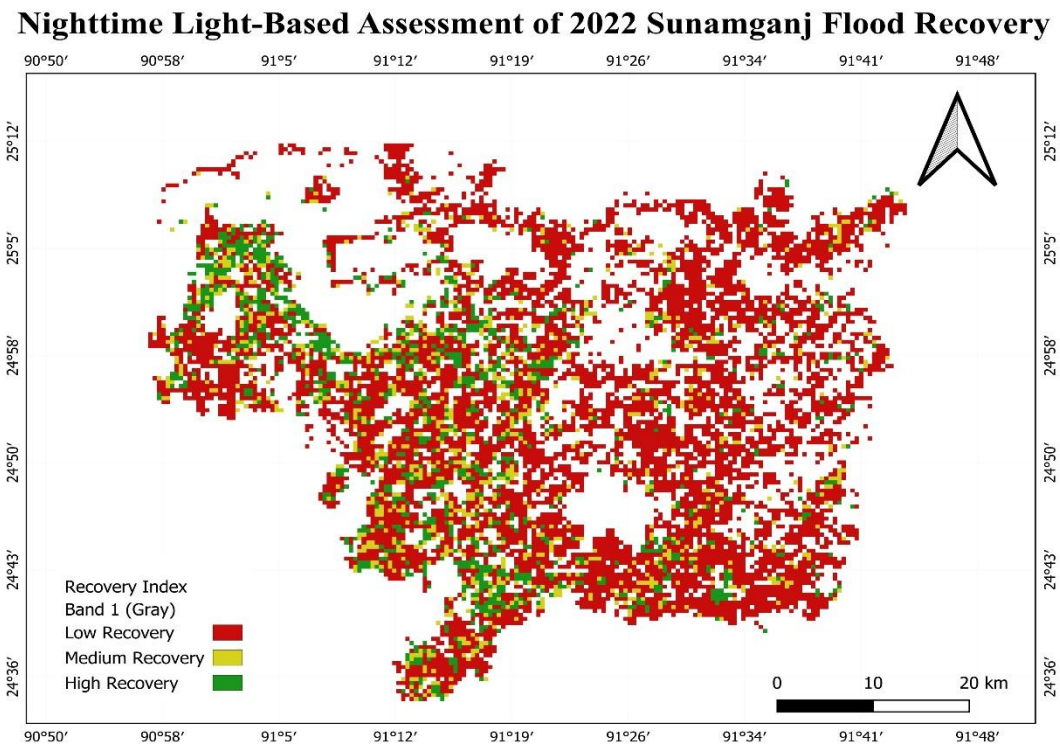

Figure 4: Nighttime Light-Based Assessment of 2022 Sunamganj Flood Recovery.

5. Discussion

The most striking finding is the **Road Density coefficient (19.86)**. In the water-logged landscape of Sunamganj, roads are typically built on raised embankments. These corridors serve as the backbone for electrical infrastructure and provide the only accessible routes for emergency repair teams.

The negative coefficients for **Wealth** and **Population Density** suggest a "vulnerability paradox." While urban areas have more resources, the complexity of their electrical grids and the density of damage to informal settlements may lead to longer restoration times compared to simpler, rural grid connections. Furthermore, the 2022 flood was so severe that it "leveled" the impact across

socio-economic classes, where even wealthy households were rendered immobile by the rising waters.

6. Conclusion

The 2022 Sunamganj flood recovery was heavily dictated by physical infrastructure. While population and wealth play roles in long-term resilience, the **immediate restoration of economic activity (as proxied by NTL)** is fundamentally tied to the road network. Policy recommendations should focus on the "flood-proofing" of embankments and road-adjacent power lines to ensure faster recovery in future climate-extreme events.

7. Limitations

- **Cloud Interference in NTL:** Despite using gap-filled data, the heavy cloud cover during the Bangladesh monsoon can introduce noise into VIIRS daily observations.
- Wealth data may not fully represent local conditions.
- Lack of ground validation data.
- Regression model does not include environmental variables such as elevation or flood depth.